

UniPulse

Software Requirements Document (SRD)

1. Introduction

UniPulse is a campus-based social and information-sharing platform designed to enable university students to interact, share updates, access learning resources, navigate campus, and communicate in real time.

2. Intended Users

- Students (Undergraduate & Postgraduate)
- Correspondents (Appointed by Admin)
- Administrators (Full system control)

3. Core Features

Common Room:

A public social feed where students can post updates, view posts chronologically, like, comment, and delete their own posts. Admins can delete any post.

Resources Page:

A digital repository of academic materials (PDFs, books, notes). Search and categorization supported.

Campus News:

Only Admins and appointed Correspondents can publish verified campus updates.

Campus Map:

Interactive campus map powered by Leaflet API. Displays faculties, hostels, halls, and key locations.

Chat System:

Real-time 1-on-1 messaging using WebSockets. Accessible via floating action button at bottom-right of screen.

Admin Dashboard:

Admins can monitor users, delete posts, promote/demote users, appoint correspondents, and suspend members.

4. Roles & Permissions

Student:

- Post in Common Room
- Chat
- View resources and news

Correspondent:

- All student permissions
- Post in Campus News

Admin:

- Full moderation control
- Appoint/remove roles
- Delete any post
- Suspend users

Special Rule:

If a user registers with the email "unipulse-admin@gmail.com", they are automatically assigned Admin role.

5. Technical Stack

Backend: Node.js (Express.js)

Database: MongoDB

Real-time Communication: Socket.io

Map Integration: Leaflet API

Authentication: JWT + bcrypt

Deployment: Cloud-ready (AWS compatible)

6. UI/UX Requirements

Desktop:

Opera-style layout with left sidebar navigation, central feed, and right information panel. Dark mode default.

Mobile:

Responsive layout with hamburger navigation menu and persistent floating chat button.

7. Non-Functional Requirements

- Secure authentication
- Role-based authorization
- Scalable to thousands of concurrent users
- Page load time under 2 seconds
- Real-time chat latency under 1 second